

Nicholas Cambre

Systems & Automation Engineer · Seattle, WA
nicholascambre0077@icloud.com · github.com/twinklingtails · nicholascambre.dev

Professional Summary

Self-taught systems and automation engineer focused on local-first AI infrastructure, self-hosted platforms, and resilient operational environments. Experienced building and maintaining multi-tenant Next.js hosting infrastructure, AI inference pipelines, Linux server environments, and automation systems spanning both software and operational workflows. Comfortable across the stack — from C++ inference systems and GPU-backed AI tooling to TypeScript frontend development and Linux infrastructure hardening. Strong preference for durable, observable systems over unnecessary complexity.

Technical Skills

Languages	TypeScript, Python, C++, JavaScript, Bash
Frameworks	Next.js, React, Node.js, Tailwind CSS, PyTorch, llama.cpp
Infrastructure	Linux, Docker, NGINX, PM2, Cloudflare, Tailscale, Home Assistant
Real-Time Systems	MQTT, Server-Sent Events, WebSockets, Node-RED, MDX
AI & Audio	Local LLM inference, Neural TTS, Whisper STT, GPU inference
Practices	Self-hosting, observability, server hardening, reverse proxy architecture

Selected Projects

Jemma AI — Local-First AI Assistant

Designed and developed a local-first AI assistant platform using C++, llama.cpp, real-time streaming infrastructure, neural voice synthesis, and self-hosted deployment tooling.

Managed Self-Hosted Web Platform

Built and currently operate a multi-tenant hosting platform supporting production Next.js sites with reverse proxy infrastructure, SSL management, and deployment orchestration.

Jemma Voice Pipeline

Developed a local neural text-to-speech system with GPU acceleration and low-latency inference averaging approximately 23ms per generated sentence on RTX 2080 Ti hardware.

Cloudflare + NGINX Edge Infrastructure

Engineered secure self-hosted edge routing infrastructure exposing internal services while maintaining operational control of backend systems.

Professional Experience

Inventory & Operations Systems Specialist

Seattle, WA · 2025 – Present

Support operational systems, inventory control, and process improvement initiatives within a medical device manufacturing environment. Collaborate cross-functionally on workflow optimization, operational visibility, and scalable process improvements.

Education

Associate's Degree — In Progress

Planned continuation into Bachelor's coursework following completion.

Independent Technical Education

Developed technical skills through self-directed infrastructure projects, source-code analysis, Linux systems administration, AI-assisted development workflows, and hands-on debugging.

Additional Information

Interested in systems engineering, infrastructure automation, AI tooling, and local-first computing environments. Passionate about building observable, maintainable systems that prioritize reliability over unnecessary complexity.